LESSON 02 · MISSION GUIDE

The Loot Split

Teach Python to make decisions, and to share out the loot without copy-paste.

From	**Captain Siv, quest master**
To	**Jen, Keeper of the Guild Ledger**
By raven	**The Queen's purse (the napkin)**

Keeper. The Queen has written the rules for her royal purse on a napkin at the feast. An actual napkin. It is pinned to the notice board with a dagger: nothing for novices with fewer than 3 quests done, a quarter of the reward on top for difficulty 4 or 5, a tenth for a 3, nothing below.

Simple enough for a person. Not for the Grimoire, which the last time it ran paid everyone, including Biscuit, and produced a total of minus forty crowns. Dalvric called it "a rounding thing". I need the same decision for every quest on the board, every week, and a total that is the total. Orrin says that's called "a loop". Yes or no will do. — Siv

What you'll be able to do after today

- [ ] Make a script choose between outcomes with `if`, `elif` and `else`
- [ ] Repeat a block for every item with `for`, and a fixed number of times with `range()`
- [ ] Repeat until something is true with `while`, and stop a runaway loop
- [ ] Keep a running total, a counter and a "best so far" across a loop
- [ ] Skip an item with `continue`, and read indentation the way Python reads it

Why this matters at work

- **Every rule is an `if`.** Purse rules, reward bands, who may take which quest: "if this, then that, otherwise...". As code, the rule is applied identically to every quest.
- **"Drag the formula down" is a loop.** It doesn't care whether there are 5 rows or 50,000, and it can't be dragged one row short.

Four big ideas

1. Indentation is the structure	VBA has `End If` and `Next`. Python has a colon and 4 spaces: the indented lines belong to the `if` or `for`. Un-indent, and you're outside it.
2. First true wins	In an `if / elif / else` chain Python runs the *first* true branch and skips the rest. Put the strictest check first.
3. `for` hands you one item at a time	`for hero in heroes:` is `For Each`. One item per trip; you never write `i = i + 1`.
4. Set up before, finish after	A running total starts at 0 *before* the loop, grows *inside*, is printed *after*. Put the `= 0` inside and it resets every trip (ask Dalvric).

Three habits that will bite you

- Forgetting the **colon** at the end of `if`, `elif`, `else`, `for`, `while` lines.
- Writing `=` in a condition. `if difficulty = 3:` is a syntax error; you mean `==`.
- A `while` loop whose condition never becomes false. `Ctrl+C` stops it. Then fix the loop.

How today works

15 MIN
Lecture
Lars runs the napkin rules live, then loops them

60 MIN
Exercise sheet
Seven ravens. You write every file yourself

15 MIN
Grimoire, page 2
The purse run that paid Biscuit

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Captain Siv	One purse decision	`if / elif / else`
2	Captain Siv	The purse run, for every leader	`for`, running total
3	Pell	Twelve days of supplies	`range()`, running total
4	The Queen	Reward bands	`elif` chain, counters, "highest so far"
5	Orrin	How many months until the Wyrm bounty passes 3,500?	`while`
6	Pell	Free rations	`continue`, `+=`
7 ⭐	Pell	Dalvric's guild calendar	everything, plus `%`

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if you're feeling brave), each printing the expected output
- [ ] Dalvric's purse run pays the right people and the total is 1,140.00 cr, with one commit per bug
- [ ] Everything is pushed to `lesson-02` and you've opened a pull request



Next lesson: Dalvric's Tangle. You'll notice today that you typed the purse rule twice. Dalvric typed it forty-one times. Next time you'll write it once, give it a name, and call it whenever you like: that's a function.